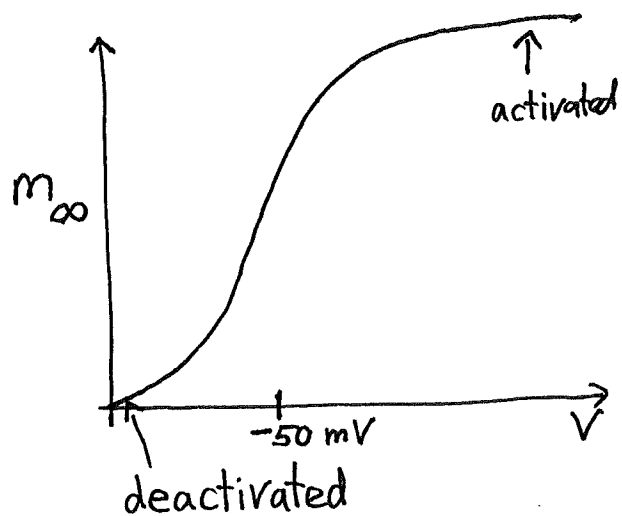
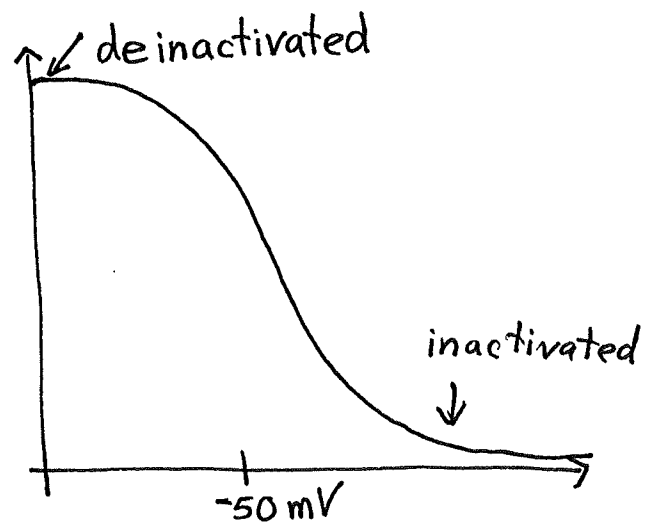


Sodium "activates" at high V



Sodium "inactivates" at high V



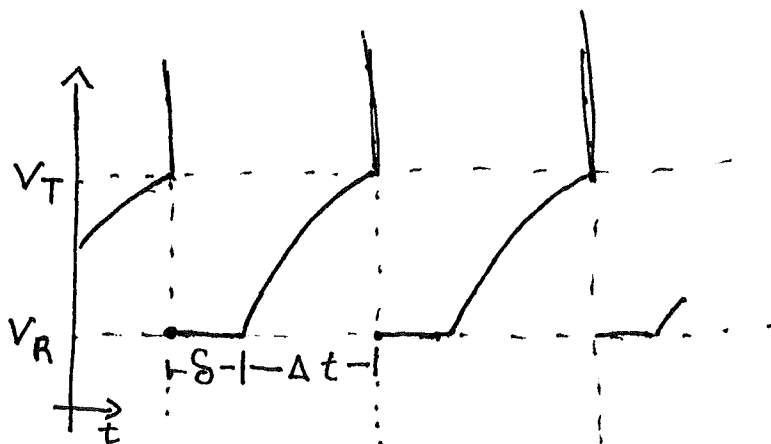
Potassium similarly activates at high V , but it does not inactivate.

Refractory period:

Sodium de-inactivation takes time, so action potentials can't be too closely spaced. F-I curves thus saturate at high currents.

Refractory periods integrate-and-fire models:

$$\lambda = \frac{1}{\delta + \Delta t} \quad ; \quad \delta: \text{refractory period} \\ \Delta t: \text{derived earlier}$$



$$\lim_{I_0 \rightarrow \infty} \lambda = \frac{1}{\delta}$$

Important remarks:

- Membrane capacitance & conductances scale with membrane area. It's convenient to normalize.

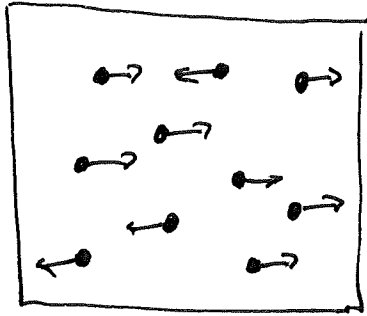
$$C = \frac{C}{A}, g_L = \frac{G_L}{A}, \text{ etc.}$$

- Can account for neuron morphology: cable equation; multicompartment modeling
- Primacy of action potentials: subthreshold signals attenuate, action potentials propagate
- Cellular diversity: different cells have different channels and morphologies. The Hodgkin-Huxley framework carries over.

Neural coding:

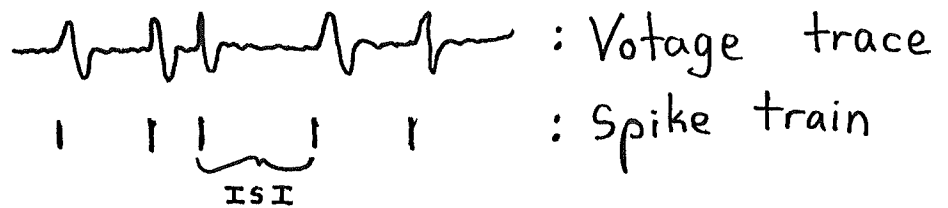
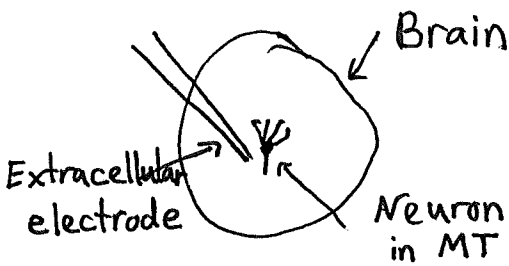
How does brain activity relate to the external world, cognition, and behavior?

Random dot kinetogram: (canonical example)

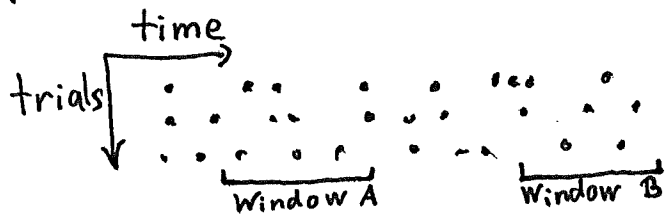


"Rightward" random dot motion
Motion strength controlled by coherence
Can vary direction of motion
Pioneered by Shadlen & Newsome

In vivo electrophysiology:



Repeat exact stimulus many times:



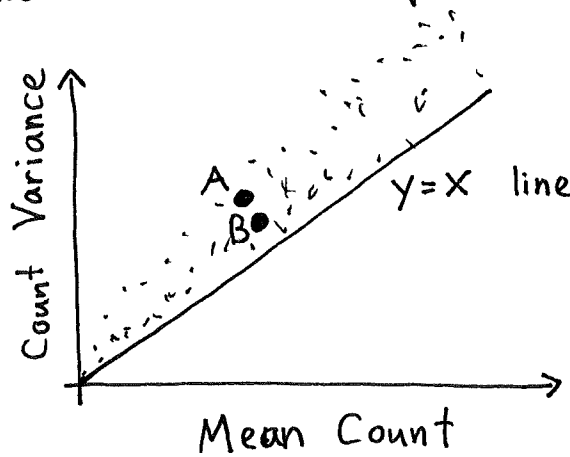
Raster plot



Peristimulus time histogram (PSTH)

What is the trial-to-trial variability?

Method 1: Count spikes



$$y=x \text{ line ; Fano Factor} = \frac{\text{Variance}}{\text{Mean}} \approx 1$$

$$P(k) = \frac{\lambda^k e^{-\lambda}}{k!} ;$$

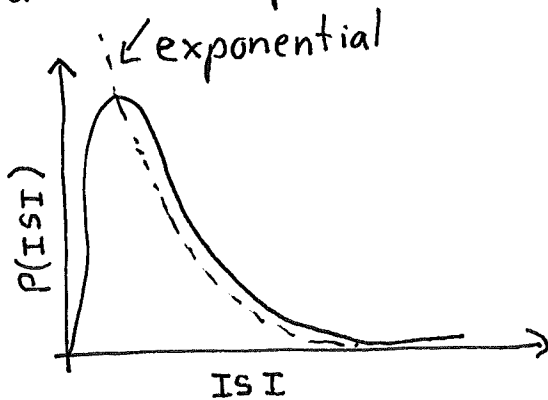
Poisson distribution:

λ : mean spike count

k : observed spike count

$$\lambda = \underbrace{r}_{\text{Firing rate}} \underbrace{\Delta t}_{\text{Bin size}}$$

Method 2: Compute inter-spike interval (ISI)



$$\text{Coefficient of variation} = \frac{\sigma}{\mu} \approx 1$$

$$P(t_{\text{ISI}}) = \frac{1}{\tau} e^{-t_{\text{ISI}}/\tau} ;$$

Exponential distribution:

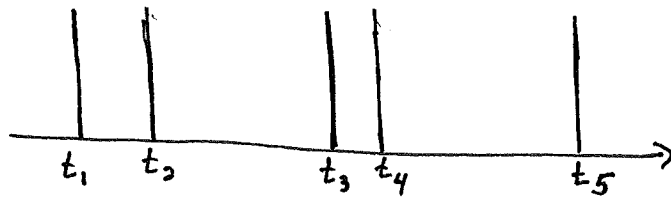
τ : mean ISI

t_{ISI} : ISI value, $t_{\text{ISI}} > 0$

$$\tau = 1/r$$

NOTE: $P(\text{ISI})$ is very different that the regular spiking of typical neuron models, such as LIF with constant drive.

Poisson point process:
Statistical model of spike train



The spike probability is uniform in time: firing rate
is fundamental quantity

Generates Poisson counts & exponential ISIs
Spike times are continuous

The problem of neural encoding:

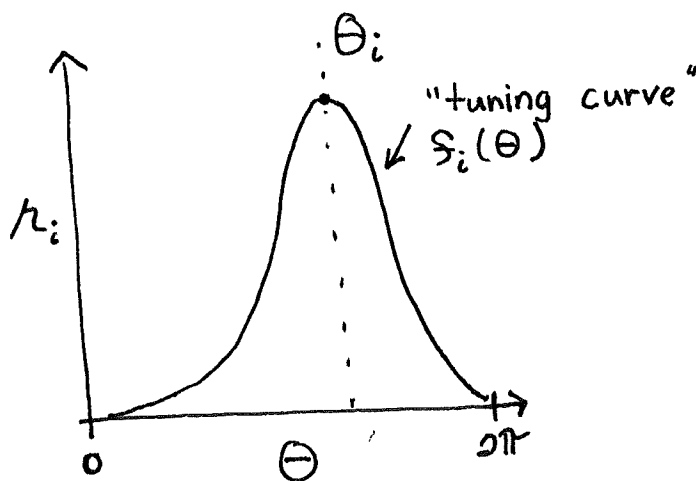
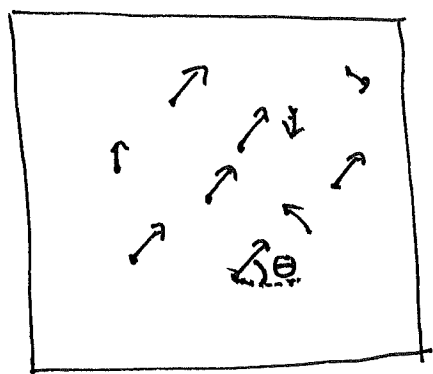
First guess: Find response as a function of stimulus

$$\underbrace{n}_{\text{neural activity}} = \underbrace{n(s)}_{\text{stimulus}}$$

Better guess: Find response probability as a function of stimulus

$$\underbrace{n}_{\text{"is drawn from"}} \sim \underbrace{P(n|s)}_{\text{Probability of response given stimulus}}$$

Example: Encoding of motion direction by MT neuron



Point process model:
$$P_i(k|\theta) = \frac{(s_i(\theta)\Delta t)^k e^{-s_i(\theta)\Delta t}}{k!}$$

For a population of N independent neurons

$$\underbrace{P(k_1, k_2, \dots, k_N | \theta)}_{\text{Spike counts from each neuron}} = \prod_{i=1}^N \underbrace{P_i(k_i | \theta)}_{\text{single neuron statistics}}$$

How accurately can we decode stimuli from neural activity?

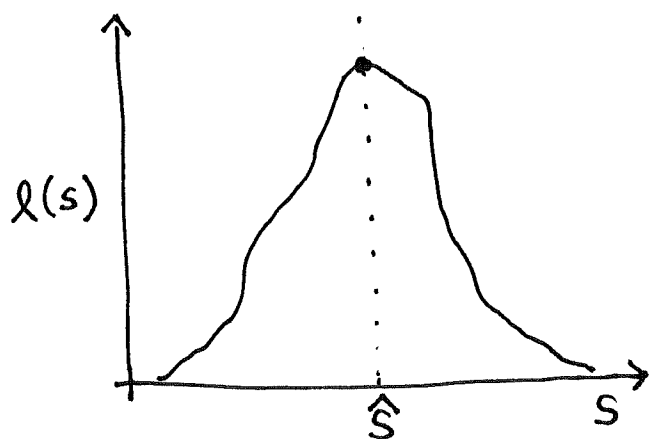
Maximum likelihood decoding:

$$L(s) = P(n|s)$$

$$\ell(s) = \log P(n|s)$$

usually base e

$$\hat{s} = \arg \max_s L(s) = \arg \max_s \ell(s)$$



At maximum:

$$\frac{d\ell(\hat{s})}{ds} = 0 ; \frac{d^2\ell(\hat{s})}{ds^2} \leq 0$$

Want this to be highly negative

Asymptotically (i.e., given lots of data, many neurons), the maximum likelihood estimate is normally distributed.

$$\hat{s} \sim \mathcal{N}(\underbrace{s_0}_{\text{True stimulus}}, 1/J(s_0))$$

$$J(s) = - \left\langle \frac{d^2\ell(s)}{ds^2} \right\rangle_{n|s}$$

Fisher information High curvature of $\ell(s)$
gives low variance estimates